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ARTICLE

Controls on variations of platinum-group element concentrations in the sulfide ores of the Jinchuan Ni-Cu deposit, western China

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Abstract Ongoing underground exploration in the giant Jinchuan Ni-Cu sulfide deposit in western China is beginning to emphasize the potential for Cu-, Pt-, and Pd-rich sulfide ores that may have formed by sulfide liquid fractionation. The success of such an effort relies on whether or not fractional crystallization of sulfide occurred in the Jinchuan system. In this paper, we used available PGE data to evaluate such a process. We found that about two thirds of the 126 samples analyzed to date exhibit significant decoupling not only between Pt and Pd but also between Ru, Rh, and Ir. The best explanation for the decoupling is postmagmatic hydrothermal alteration, which affected not only silicates but also sulfides. The effects of postmagmatic alteration must be considered when using metal and isotopic ratios to evaluate primary mineralization. PGE variations in the remaining one third of the samples with Ir/(Ir + Ru) = 0.3–0.7, Ir/(Ir + Rh) = 0.4–0.8, and Pt/(Pt + Pd) = 0.3–0.7 indicate variable R-factors within individual ore bodies as well as the entire deposit, consistent with the interpretation that multiple sulfide-bearing magmas from depth were involved in the formation of the Jinchuan deposit. The mantle-normalized PGE patterns of the least-altered samples from the Jinchuan deposit are similar to the picrite-related Pechenga Ni-Cu sulfide deposit in Russia. PGE variations that can be related to sulfide liquid fractionation are observed in orebody-1 and orebody-24 but not in orebody-2 at Jinchuan. Exploration for Cu-, Pt-, and Pd-rich sulfide ores that may have been expelled into fractures in the footwalls of orebody-1 and orebody-24 appears to be justified.

Keywords Platinum · Iridium · Nickel · Sulfide · Jinchuan

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Introduction

The Jinchuan Ni-Cu deposit in China is one of the largest magmatic sulfide deposits in the world. It contains >500 million metric tons of sulfide ores with grades of 1.1 wt% Ni and 0.7 wt% Cu (Chai and Naldrett 1992a). The Jinchuan deposit has been mined primarily for Ni and Cu since 1964. Platinum and Pd have became more valuable by-products in recent years due to the dramatic increase in the prices of these metals. The annual output of Pt + Pd from Jinchuan now accounts for ~95% of the total production in China. Inspired by examples from other major magmatic sulfide deposits in the world such as the Sudbury Ni-Cu deposits in Canada (Li et al. 1992) and the Talnakh Ni-Cu deposits in Siberia (Zientek et al. 1994), underground exploration at Jinchuan has recently begun to focus on the potential for Cu-, Pt-, and Pd-rich sulfide ores that formed by sulfide liquid fractionation. To assist this endeavor, we have analyzed 32 sulfide-bearing samples collected from current underground drifts in three different ore bodies for the concentrations of platinum-group elements (PGE). We use our data, together with data from previous studies by Chai and Naldrett (1992a), Song et al. (2006) and Yang et al. (2006), to evaluate fundamental controls on the compositional variations of sulfide ores in the Jinchuan deposit. We use element correlation and numerical modeling to assess the roles of R-factor (silicate liquid to sulfide liquid ratio during sulfide segregation), fractional crystallization of monosulfide solid solution (mss) from sulfide liquid and hydrothermal modification in the final distributions of PGE in the Jinchuan deposit. Our results bear important implications for the ongoing underground exploration.

Geological background

The Jinchuan Ni–Cu deposit is hosted by an ultramafic body, which occurs along the southwestern margin of the Sino-Korean craton (also referred to as the North China craton) (Fig. 1a). Immediate country rocks to the intrusion are marble, gneisses, and granitic migmatites that belong to the Proterozoic Baijiazuizi Formation. Both the age and metamorphic grade of the Proterozoic sequence decrease to the southwest (Chai and Naldrett 1992b). Rocks of the uppermost Proterozoic unit include sericite schist, carbonate schist, limestone, and intercalated andersitic metavolcanics. In the Jinchuan area, the northern boundary of the Proterozoic sequences is a fault contact with post-Palaeozoic sedimentary rocks (Fig. 1b).

The surface exposure of the Jinchuan intrusion measures ~6,500m long and ~500m wide (Fig. 1b). The vertical downward extension of the intrusion exceeds 1,100m in its central part (Fig. 1c). Harzburgite, lherzolite, and dunite are the major rock types of the intrusion. Recent SHRIMP U-Pb ages of baddeleyite and zircon from lherzolites by Li et al. (2005) are 812 ± 26 Ma and 827 ± 8 Ma. The SHRIMP U-Pb age of zircon from the dolerite dykes that cut the Jinchuan intrusion is 828 ± 8 Ma (Li et al. 2005). Based on these ages, Li et al. (2005) postulated that the Jinchuan intrusion was related to the mantle plume activity that triggered the breakup of Rodinia.

The Jinchuan intrusion is divided into three segments by a series of northeast-trending strike-slip faults, namely, the western, central, and eastern segments (Fig. 1b). They host the three most important sulfide ore bodies at Jinchuan, namely, orebody-24, orebody-1, and orebody-2, respectively (Fig. 1c). The sulfides mainly occur in the form of disseminated and net-textured pyrrhotite, chalcopyrite, and pentlandite, which commonly comprise between 2 and 30 vol.% of the olivine-rich rocks. Massive sulfide mineralization is less common. It occurs as small concordant lenses or discordant veins. Sulfide assemblages in the massive ores are highly variable; ranging from pyrrhotite dominant to chalcopyrite dominant. Rare magnetite-rich massive ores are also present.

Chai and Naldrett (1992b) proposed that the Jinchuan intrusion represented the root zone of a much larger layered intrusion and the central part of the intrusion was a subvertical feeder to the magma chamber. In contrast, Tang (1993) suggested that the intrusion formed by multiple inputs from a deep-seated stratified magma chamber. Sulfide-bearing magma was proposed to have been extracted from the lower portion of the deep chamber, whereas sulfide-poor intrusions in the area were extracted from the upper portion. De Waal et al. (2004) and Li et al. (2004) showed via olivine compositions, whole-rock trace element ratios and sulfide ore compositions that multiple pulses of magma were involved in the formation of the Jinchuan deposit. Like Tang (1993), de Waal et al. (2004) and Song et al. (2006) called upon emplacement of sulfide-bearing and sulfide-free pulses from a deep, stratified magma chamber to explain the occurrence of sulfide-barren as well as sulfide-bearing rocks in the Jinchuan area.

Different opinions exist with respect to the original geometry of the Jinchuan intrusion. Chai and Naldrett (1992b) proposed that the Jinchuan intrusion was emplaced as a near-vertical dyke based on the sub-horizontal layering structures in the eastern part of the intrusion. In contrast, de Waal et al. (2004) suggested that the Jinchuan intrusion was emplaced as a sill based on similar dipping angles of internal lithological boundaries and the footwall contacts. Lehmann et al. (2007) used the overall concordant relation between the northern contact and the country rock strata in the central part of the intrusion to support the sill model. The debate will continue until data from detailed textural analysis of fresh rocks in the Jinchuan intrusion become available. Unfortunately, fresh rocks in the Jinchuan intrusion are rare due to significant postmagmatic hydrothermal alteration and deformation.

Chai and Naldrett (1992b) were the first to recognize that the parental magma of the Jinchuan intrusion was high-MgO basalt instead of ultramafic magma, which many Chinese geologists initially believed. Lehmann et al. (2007) suggested that the MgO content in the primary magma of the Jinchuan intrusion could be higher than that proposed by Chai and Naldrett (1992b) because the parental magma of the Jinchuan intrusion may have been contaminated by continental crust at depth. Li et al. (2005) suggested that the parental magma of the Jinchuan intrusion was derived from the subcontinental lithospheric mantle and experienced crustal contamination based on negative εNd (T = 825Ma) values (−8.9 to −12.0) that decrease with increasing La/Sm. Song et al. (2006) and Lehmann et al. (2007) reported negative Nb and Ta anomalies in the Jinchuan rocks and attributed these anomalies to contamination with granitoid materials. Song et al. (2006) proposed that sulfide saturation was achieved at depth as a result of fractional crystallization as well as crustal contamination. In contrast,

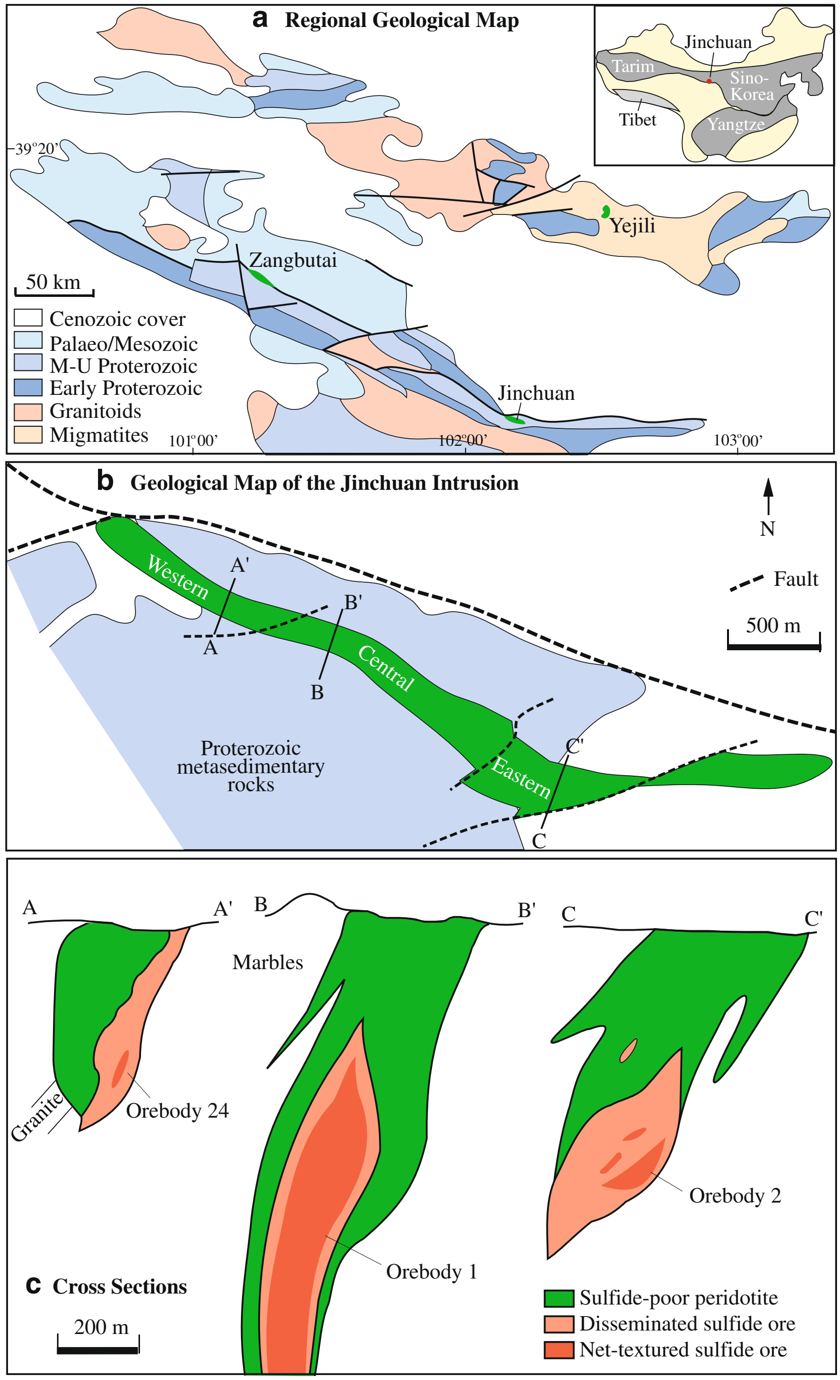

Fig. 1 **a** Location and regional geologic map of the Jinchuan area. **b** Plan view of the Jinchuan intrusion. **c** Cross-sections based on drilling. Modified from Chai and Naldrett (1992a) and Li et al. (2004)

Lehmann et al. (2007) suggested that sulfide saturation was induced by the addition of carbonate-rich fluids which increased the oxygen fugacity of the magma and hereby decreased the maximum solubility of sulfur in the magma.

Postmagmatic hydrothermal alteration and/or metamorphism that has affected the Jinchuan intrusion and sulfide ores was recognized by Chai and Naldrett (1992b), Barnes and Tang (1999), de Waal et al. (2004), Li et al. (2004), Yang et al. (2006), Ripley et al. (2005), and Lehmann et al. (2007). The most detailed account of alteration in the Jinchuan intrusion is given by Ripley et al. (2005). Olivine is in various states of replacement by serpentine, magnetite, chlorite, and actinolite, and interstitial pyroxene and plagioclase have been locally converted to chlorite, tremolite, epidote, and more sodic plagioclase. In many places, the contact zones of the intrusion have been sheared and almost completely altered to tremolite and chlorite. Secondary calcite veins with minor pyrite are common in the sheared zones. Locally as much as 30 vol.% of the magmatic sulfide assemblage in the Jinchuan intrusion has been replaced by secondary silicates and oxide minerals, with an average of 10 vol.%. Element redistribution due to postmagmatic hydrothermal alteration in the Jinchuan intrusion has created a problem for proper interpretation of whole-rock PGE data (Yang et al. 2006), stable isotopes (Ripley et al. 2005), and Re–Os systematics (Yang et al. 2008).

Samples and analytical methods

Samples from this study were collected from underground drifts at an elevation of 1,220m in orebody-24, and 1,118m in orebody-1 and orebody-2. The corresponding depths from the surface are 510 and 612m, respectively.

The compositions of platinum-group minerals (PGM) were determined by wavelength-dispersive X-ray analysis using a CAMECA SX50 electron microprobe at Indiana University. The analytical conditions were 15kV, 20nA beam current, 1-μm beam size and peak-counting time of 20s. Synthetic FeS, NiS, and InAs, and pure Pt, Pd, Bi, and Te standards were used for calibration. Raw data were corrected using the PAP program supplied by CAMECA.

The contents of S in whole-rock samples were determined by LECO at China University of Geosciences in Beijing. Whole-rock Ni, Cu, and Co contents were analyzed by inductively coupled plasma-mass spectrometry (ICP-MS) at the University of Hong Kong. The procedures, precision, and accuracy are given in Qi et al. (2000). Whole-rock PGE contents were determined using the combination of sodium peroxide fusion, tellurium co-precipitation, and ICP-MS at the University of Hong Kong. The procedures, precision and accuracy are given in Qi et al. (2004). The PGE contents in the blanks are Ir < 0.15ppb, Ru < 0.015ppb, Rh < 0.05ppb, Pt < 0.5ppb, and Pd < 0.5ppb.

Results

Platinum group minerals

Platinum group minerals (PGM) in our samples are sperrylite ($PtAs_2$), froodite ($PdBi_2$), and michenerite (PdBiTe). In addition to these PGM, Yang et al. (2006) have reported moncheite ($PtTe_2$), merenskyite ($PdTe_2$), kotulskite (PdTe), and Ag–Pd–Te–Bi compounds in the Jinchuan deposit. In our samples, sperrylite occurs mainly as euhedral crystals with sizes varying from 20 to 100 μm in diameter. They are closely associated with base metal sulfides (Fig. 2a) or hydrous silicates (Fig. 2b). The compositions of the sperrylite crystals analyzed are close to $Pt_{33.8}$ $Fe_{0.27}$ $Ni_{0.03}$ $As_{65.5}$ $S_{0.39}Te_{0.05}$. Michenerite occurs as subhedral crystals with other PGM including froodite and other Te–Bi compounds, such as BiNi, in the margins of base metal sulfide assemblages or with secondary silicates such as chlorite or serpentine (Fig. 2c). The compositions of michenerite crystals are close to $Pd_{31.7}Fe_{0.94}Ni_{0.44}Te_{35.9}Bi_{27.4}$ $S_{2.8}As_{0.9}$.

Element correlations in whole rocks

The concentrations of PGE, Cu, Co, Ni, and S in our samples are listed in Table 1. Data from the previous studies by Chai and Naldrett (1992a), Song et al. (2006) and Yang et al. (2006) are included in the following comparisons. All of the data together indicate that there is a good correlation between Ni and S contents, particularly for the samples containing <15wt% S (Fig. 3a). The samples with <15wt% S are disseminated and net-textured sulfides whereas the samples with >28wt% are massive sulfides. The contents of Ni in the massive sulfides are more scattered, indicating a more heterogeneous distribution of pentlandite in the massive sulfides than the disseminated and net-textured sulfides. Correlation between Cu and S is poor for all samples (Fig. 3b), indicating a heterogeneous distribution of chalcopyrite in the samples. There is no

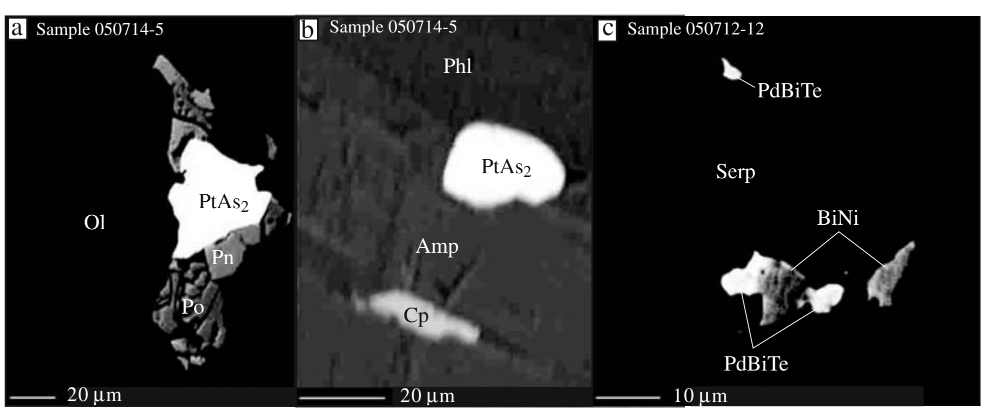

Fig. 2 Back-scattered electron images of platinum-group minerals from the Jinchuan deposit. **a** Sperrylite ($PtAs_2$) occurs as an integral part of a base metal sulfide assemblage. **b** Sperrylite occurs along the grain boundaries of hydrous silicates. **c** Michenerite (PdBiTe) with a BiNi phase disseminated in a secondary silicate matrix. *Amp* amphibole, *Cp* chalcopyrite, *Phl* phlogopite, *Ol* olivine, *Serp* serpentine

Table 1 Concentrations of metals in sulfide-bearing samples from the Jinchuan Ni–Cu deposit

Sample	Ir ppb	Ru ppb	Rh ppb	Pt ppb	Pd ppb	Cu wt%	Co wt%	Ni wt%	S wt%
Orebody 24									
050709-1	1.94	4.57	1.73	50.62	38.63	0.06	0.013	0.18	0.56
050712-1	0.28	1.31	0.07	10214.00	1423.16	2.46	0.041	3.15	8.36
050712-2	2.70	8.56	3.93	1976.40	448.21	3.60	0.039	1.83	10.59
050712-3	4.08	2.79	6.21	1074.11	553.84	2.80	0.050	3.05	11.07
050712-5	0.21	4.69	0.14	376.50	280.82	0.57	0.019	0.79	2.13
050712-6	0.70	0.95	0.59	450.25	207.56	0.90	0.022	0.73	2.09
050712-9	1.96	1.07	1.25	129.03	358.14	1.29	0.058	4.36	10.51
050712-11	0.84	4.20	0.09	921.78	651.28	3.75	0.023	1.21	8.86
050712-12	0.45	2.40	0.12	386.12	3295.94	3.14	0.027	1.26	7.48
050712-16	3.50	42.48	12.88	281.32	317.38	0.55	0.036	1.28	5.16
Orebody 1									
050713-1a	5.02	4.19	2.97	83.52	528.54	2.26	0.038	1.68	8.14
050713-1b	4.00	3.37	3.19	90.12	532.58	2.26	0.038	1.68	8.14
050713-2	8.24	7.36	3.51	12.99	269.72	1.77	0.040	1.91	7.67
050713-3	16.68	12.10	8.57	21.01	359.20	1.39	0.049	2.41	8.48
050713-4	18.42	20.38	21.65	960.29	253.71	0.52	0.043	1.73	7.23
050713-5	20.62	36.90	22.88	57.07	183.11	0.61	0.045	1.70	8.83
050713-6	4.49	75.12	23.04	10.22	134.78	0.85	0.046	1.84	7.80
050713-7	1.39	16.42	5.09	40.36	55.66	0.45	0.022	0.53	2.28
050713-8	2.11	34.79	10.74	30.86	70.56	0.37	0.029	0.95	3.59
050713-9	0.22	1.79	0.12	2.61	4.63	0.01	0.012	0.10	0.21
050714-1	1.53	73.96	27.49	78.94	128.96	1.74	0.037	1.22	7.42
050714-5	9.21	13.41	7.29	206.16	43.50	0.19	0.021	0.63	2.62
050714-6	2.57	14.24	5.14	225.14	51.96	0.14	0.021	0.55	3.47
050714-7	0.79	1.89	0.51	9.80	12.88	0.13	0.018	0.27	0.91
050714-9	0.48	1.96	0.39	26.50	19.18	0.03	0.012	0.17	0.45
Orebody 2									
050715-1	4.69	5.38	1.39	0.82	15.67	0.56	0.026	0.71	4.25
050715-2	0.47	0.42	0.17	20.93	23.30	0.22	0.021	0.44	1.83
050715-3	0.69	2.43	0.82	11.33	25.81	0.30	0.020	0.47	2.01
JC02-4	100.00	125.00	54.30	16.60	208.00	3.20	0.080	8.25	35.5
JC02-3	60.90	54.60	24.80	24.60	170.5	1.45	0.080	1.93	7.42
JC02-5	22.90	11.30	17.80	12.19	66.60	0.67	0.070	1.90	9.53
JC02-6	2.15	2.68	1.13	73.10	23.80	0.33	0.040	1.10	6.22
JC02-7	0.35	0.55	0.21	2.28	1.44	0.07	0.020	0.48	2.22

Sample numbers ending with *a* and *b* are duplicates.

correlation at all between Ir and S (Fig. 3c) and between Pd and S (Fig. 3d).

Element correlations in 100% sulfide

We have used the equation given by Barnes and Lightfoot (2005) (their equation 1) to calculate the concentrations of elements in 100% sulfide. The equation of Barnes and Lightfoot (2005) was formulated based on the assumption that sulfides in the rocks are present as pyrrhotite, pentlandite, and chalcopyrite, which was originally proposed by Naldrett and Duke (1980). The contents of S, Cu, and Ni in whole rocks are required for recalculation to 100% sulfide. Before recalculation to 100% sulfide, we made a correction for Ni from silicates (mostly in olivine) by assuming that the content of Ni in 100% silicate is 1,000ppm. This value is within the range of samples containing <0.2wt% S. The amount of total sulfide in a sulfide-bearing sample is estimated by the content of S multiplied by a factor of 2.8, which is within the range of massive sulfides. Errors associated with S analysis at low concentrations and Ni correction for S-poor samples increases with decreasing S content in the samples. In the following comparison, samples with S contents <0.5wt% are excluded to limit these errors.

Element correlations for orebody-1 are illustrated in Fig. 4a–f. There is no significant correlation between Pd and Ni (Fig. 4a), between Pt and Cu (Fig. 4b), between Pt

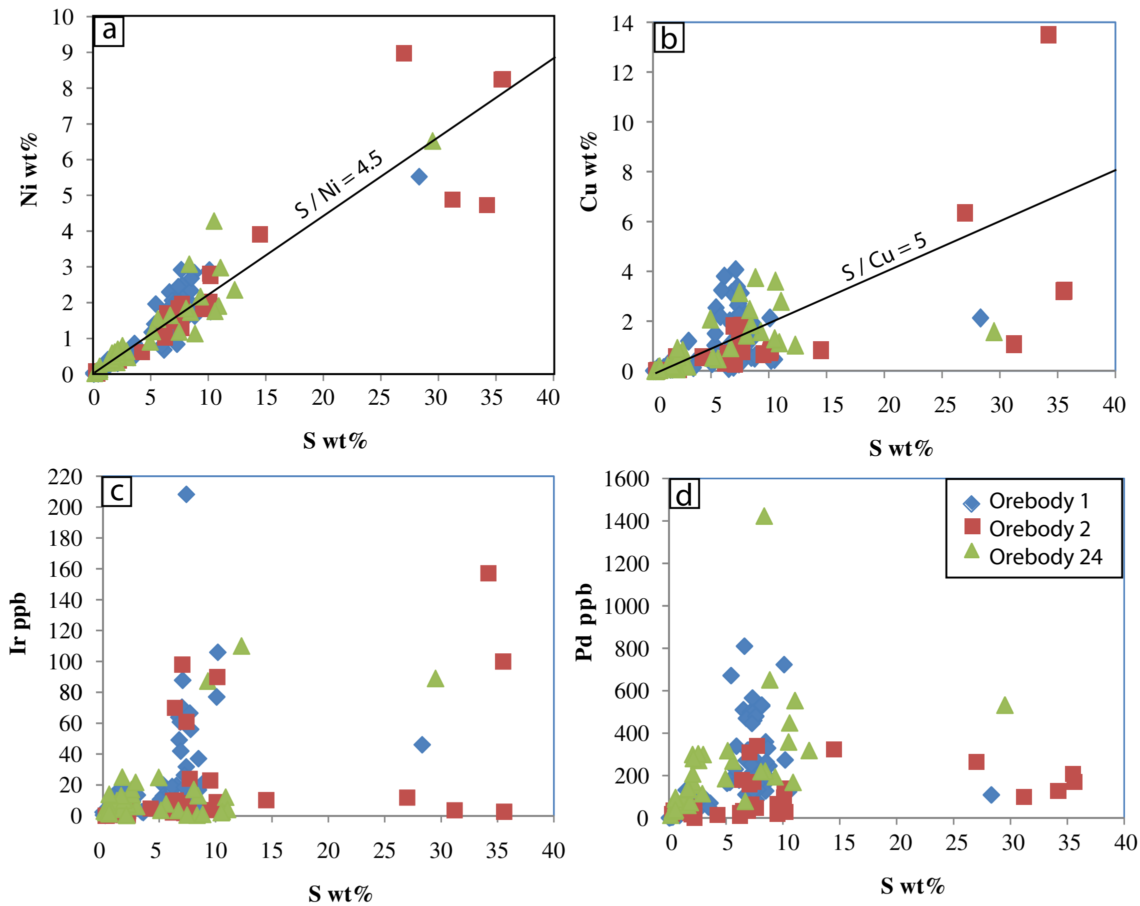

Fig. 3 Inter-elemental relations of Ni, Cu, Ir and Pd with S in whole rocks

and Ir (Fig. 4c), and between Pd and Ir (Fig. 4d). Strong positive correlations are observed between Ru and Ir (Fig. 4e), and between Rh and Ir (Fig. 4f) in the samples with Ir concentrations >70ppb. No significant correlations between Ru, Rh, and Ir are observed in the samples with Ir concentrations <70ppb (Fig. 4e,f).

In orebody-2, no significant correlation is observed between Pd and Ni for all of the samples (Fig. 5a). Four massive sulfide samples analyzed by Chai and Naldrett (1992a) show a good, positive correlation between Pd and Ni. Chai et al. (1993) concluded that in these samples, Pd was primarily hosted in pentlandite based on in situ analysis of Pd in sulfide minerals by accelerator mass spectrometry. No correlation is observed between Pt and Cu (Fig. 5b) and between Pt and Ir for disseminated and net-textured sulfides (Fig. 5b). Massive sulfides analyzed by Chai and Nadrett (1992a) show positive correlations between Pt and both Cu and Ir. The one massive sulfide sample analyze as part of this study is depleted in Pt relative to both Cu and Ir. No significant correlation is observed between Pd and Ir (Fig. 5d). Ru and Ir exhibit strong positive correlation (Fig. 5e). Rh and Ir exhibit weak positive correlation (Fig. 5f).

In orebody-24, only the suite of samples from Song et al. (2006) shows a positive correlation between Pd and Ni (Fig. 6a). All other samples from this study as well as from Chai and Naldrett (1992a) do not indicate any significant correlation between Pd and Ni. As the same methods and laboratory were used by Song et al. (2006) and us, the discrepancy of results from these two studies is more likely related to systematic sampling bias than an analytical problem. No correlations are observed between Pt and Cu (Fig. 6b), and between Pt and Ir (Fig. 6c). No significant correlation between Pd and Ir is observed for all of the samples (Fig. 6d). Samples with less than ~40ppb Ir show no significant Ru–Ir correlation; samples with more than ~40ppb Ir show a strong positive Ru–Ir correlation (Fig. 6e). Rh and Ir exhibit weak positive correlation for all of the samples (Fig. 6f).

Distributions of element ratios

The distributions of important element ratios are illustrated in Fig. 7a–d. The ratios of Ni/(Ni + Cu) exhibit an asymmetric pattern with a peak at 0.7–0.8 (Fig. 7a). More than 95% of the samples have the ratios Ni/(Ni + Cu)

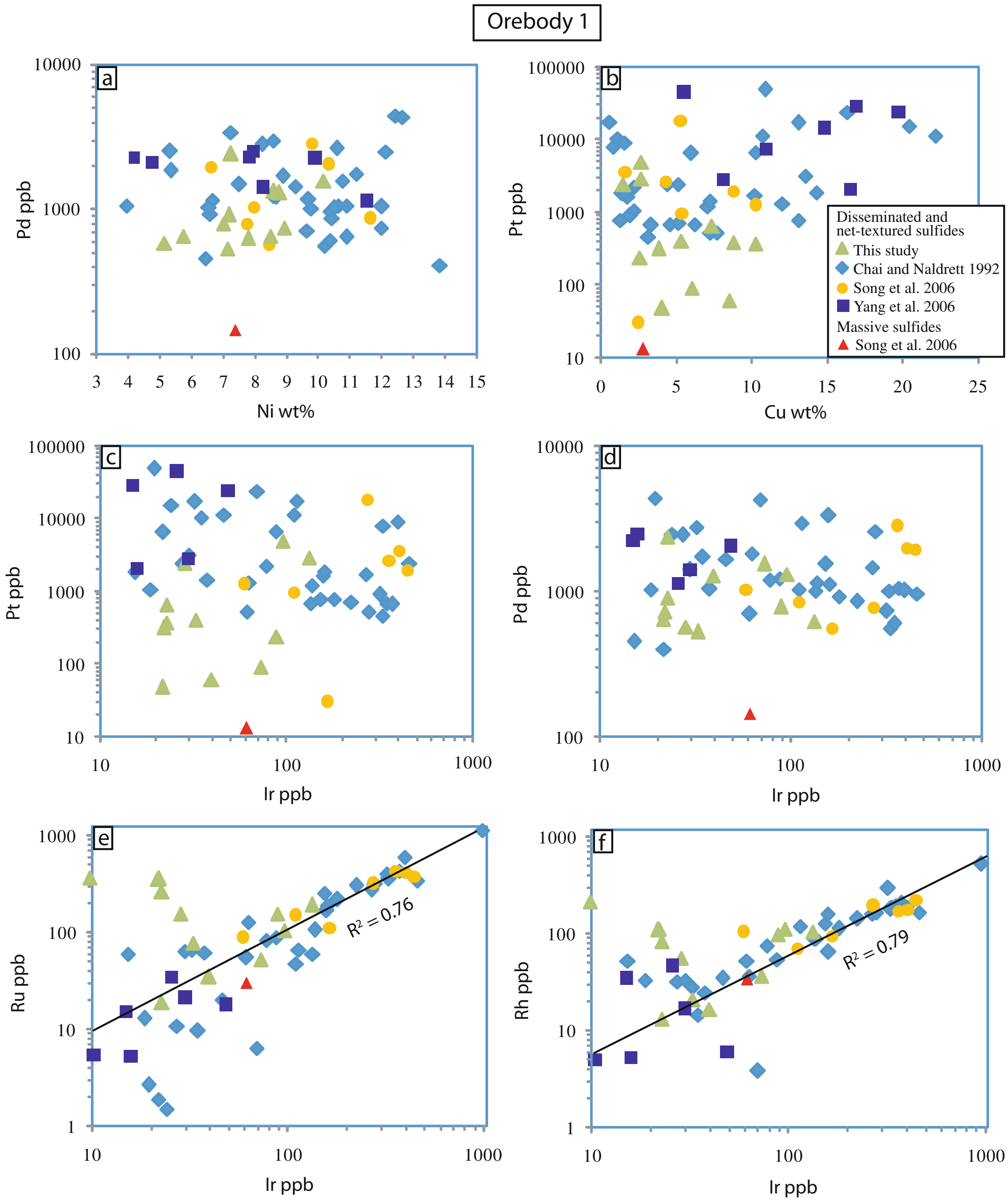

Fig. 4 Inter-elemental relations of PGE, Ni, and Cu in recalculated 100% sulfide in orebody-1. The trendlines are linear regression lines with intercept at zero for all samples. *R* is correlation coefficient

between 0.3 and 0.9. The large range of the ratio reflects highly variable modal proportions of pentlandite and chalcopyrite in the samples.

The distribution of Pt/(Pt + Pd) ratios is not as strongly asymmetric as the Ni/(Ni + Cu) ratios (Fig. 7b). It is characterized by a major peak at 0.6–0.7 and two secondary peaks at 0.1–0.2 and 0.8–0.9. The ratios of Ir/(Ir + Ru) have a nearly normal distribution pattern with a peak at 0.4–0.5 (Fig. 7c). The ratios of Ir/(Ir + Rh) have an asymmetric distribution pattern with a peak at 0.6–0.7 (Fig. 7d).

Discussion

Effect of hydrothermal alteration

Iridium, Ru and Rh have similar partition coefficients between sulfide liquid and magma, and between mss and sulfide liquid (see data complied by Barnes and Lightfoot 2005). As a result, these elements do not significantly fractionate from each other during sulfide liquid segregation from magma and subsequent mss crystallization from sulfide liquid on cool-

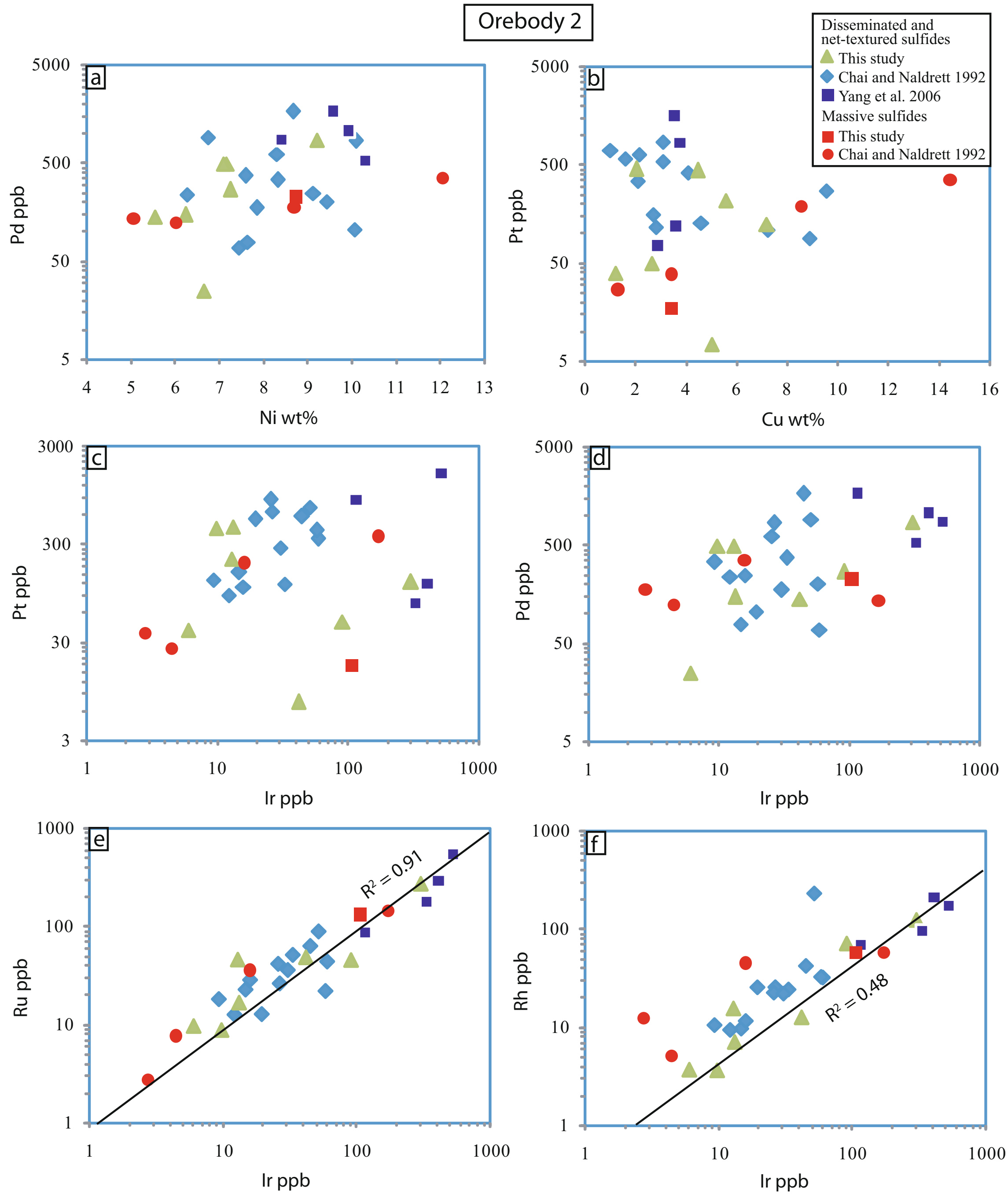

Fig. 5 Inter-elemental relations of PGE, Ni, and Cu in recalculated 100% sulfide in orebody-2. The trendlines are linear regression lines with intercept at zero for all samples. *R* is correlation coefficient

ing. The lack of positive correlation between these elements in many samples is likely due either to analytical problems at low concentrations or postmagmatic redistribution. The analytical problem is not solely responsible because samples from orebody-2 with similarly low concentrations do not exhibit significant decoupling between Ru and Ir (Fig. 5e). Furthermore, significant decoupling between Ir and Rh occurs not only at relatively low concentrations (Fig. 4f), but also at relatively high concentrations (Fig. 6f).

Similarly, Pt and Pd are not expected to significantly fractionate from each other during sulfide liquid segregation because they have similar partition coefficients between sulfide liquid and magma. These two elements also have similar partition coefficients between mss and sulfide liquid (see data compiled by Barnes and Lightfoot 2005) and they should behave similarly during mss crystallization. They may start to fractionate from each other due to the preferential partitioning of Pd into

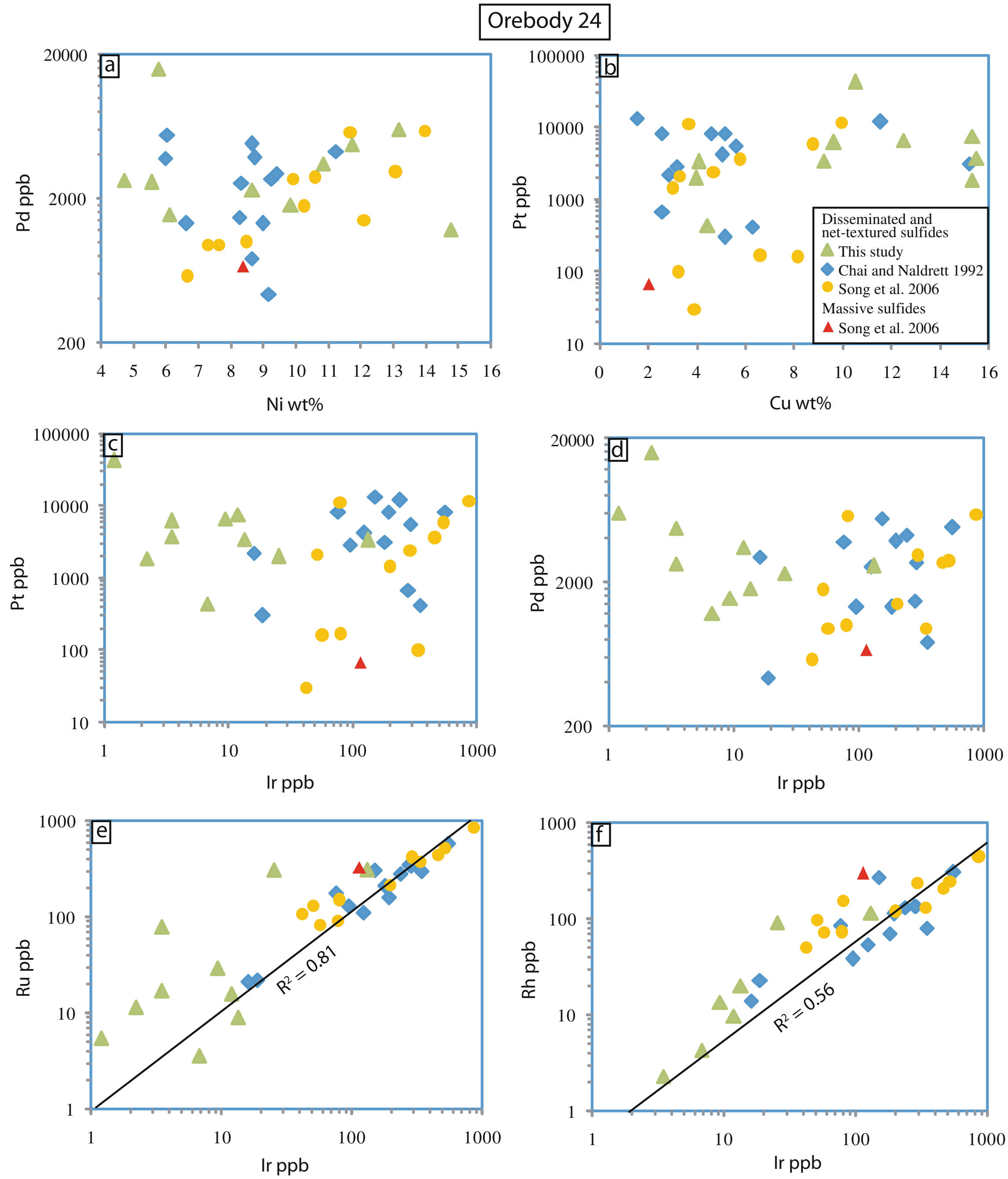

Fig. 6 Inter-elemental relations of PGE, Ni, and Cu in recalculated 100% sulfide in orebody-24. The trendlines are linear regression lines with intercept at zero for all samples. *R* is correlation coefficient

pentlandite when pentlandite starts to exsolve from mss or directly appears on the liquidus of a Cu-Ni-rich fractionated liquid. However, pentlandite exolution only takes place at temperatures below 850°C when sulfide crystallization is nearly complete (see the phase diagrams of Fe–Ni–S system complied by Naldrett 2004). Residual sulfide liquids formed at the waning stages of sulfide crystallization could be expelled into existing microfractures in the rocks, but they may be volumetrically insignificant.

The lack of positive correlation between Pd and Ni for the majority of the samples (Figs. 4a, 5a, 6a) indicates that pentlandite is not the dominant host for Pd or the concentrations of Pd in pentlandite are highly variable at sample scales. In either case, the lack of a positive Pd–Ni correlation is more consistent with hydrothermal modification than primary characteristics.

For the reasons given above, the ratios of Ir/(Ir + Ru), Ir/(Ir + Rh), and perhaps also Pt/(Pt + Pd) should be restricted

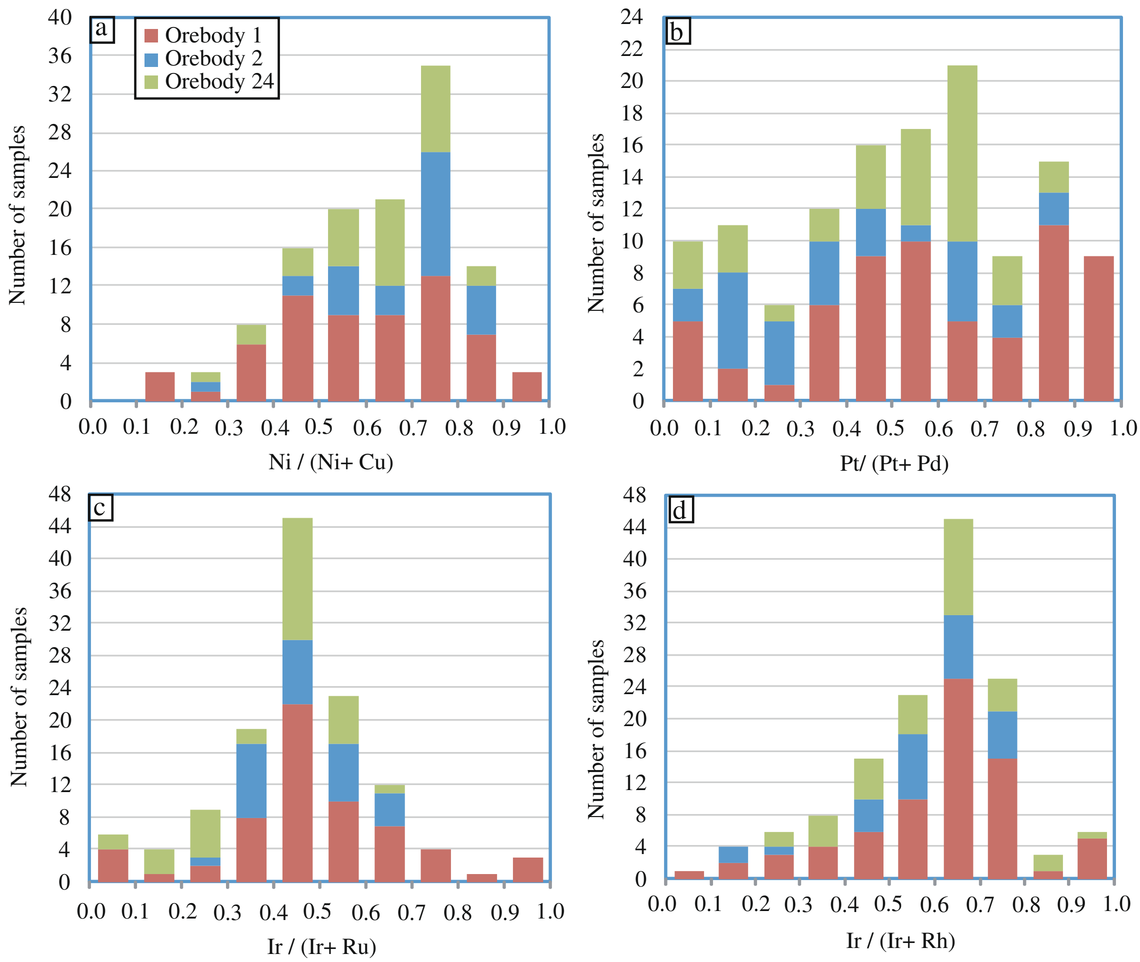

Fig. 7 Histographs showing sample distribution in terms of Ni/(Ni + Cu), Pt/(Pt + Pd), Ir/(Ir + Ru), and Ir/(Ir + Rh)

without postmagmatic hydrothermal modification. In considering the distributions of these ratios for all of the samples (Fig. 7b–d), we have used Ir/(Ir + Ru) = 0.3–0.7, Ir/(Ir + Rh) = 0.4–0.8, and Pt/(Pt + Pd) = 0.3–0.7 as cutoff values to filter out samples that may have been significantly modified by postmagmatic hydrothermal alteration. The cutoff values are arbitrary in part, but are similar to the values of typical magmatic Ni–Cu sulfide deposits at Sudbury (Naldrett et al. 1982). Among a total of 126 samples from the Jinchuan deposit, there are 46 samples left after filtering. All six massive sulfide samples were eliminated.

The filtering results are at odds with the use of massive sulfide samples from the Jinchuan deposit for Re-Os dating by Yang et al. (2008). The Ni/(Ni + Cu) ratios for the selected samples are between 0.3 and 0.9. The selected samples may represent both fresh and minimally altered magmatic sulfide ores in the Jinchuan intrusion. For the purpose of comparison in the following discussion, the metal concentrations in 100% sulfide of the selected samples are listed in Table 2. Among the 46 samples selected, 31 samples are from Chai and Naldrett (1992a), nine are from Song et al. (2006), three from Yang et al. (2006), and three samples from this study.

Roles of magmatic processes

Figure 8 illustrates the mantle-normalized PGE patterns of a few representative samples selected from Table 2. The PGE pattern of average disseminated sulfide ores (recalculated to 100% sulfide) from the Pechenga Ni-Cu deposit by Barnes et al. (2001) is included for comparison. It is seen that the medium values of PGE concentrations in the Jinchuan deposit is similar to the averages of disseminated sulfide ores from the Pechenga deposit. The sample with low Ir, Ru, and Rh from orebody-24 has a distinct pattern compared to others. It is significantly depleted in IPGE (Ir, Ru, and Rh) and enriched in PPGE (Pt and Pd), Au, and Cu compared to the Pechenga sample. These features are consistent with a residual sulfide liquid formed by mss fractional crystallization. IPGE are compatible whereas PPGE, Au, and Cu are incompatible in the mss structure. As a consequence, fractional crystallization of mss from a sulfide liquid will deplete IPGE and enrich PPGE, Au, and

Table 2 Concentrations of metals in 100% sulfide of selected samples from the Jinchuan Ni-Cu deposit[a]

Sample	Orebody #	Os ppb	Ir ppb	Ru ppb	Rh ppb	Pt ppb	Pd ppb	Au Ppb	Cu wt%	Co wt%	Ni wt%	Ref
JC-24	Orebody 1		444	389	231	2056	1972		8.7	0.0	6.6	2
JC-28	Orebody 1		109	152	72	1005	861		5.4	0.4	11.6	2
JZ-5	Orebody 1		359	443	177	2763	2847		4.3	0.0	9.8	2
JZ-6	Orebody 1		403	425	181	3780	2036		1.7	0.0	10.3	2
Y2-13	Orebody 1	34	30	21	17	2818	1409	1665	8.2	0.3	8.3	3
2W12	Orebody 1	171	158	171	161	1659	3365	510	1.8	1.5	7.2	4
2-W-6	Orebody 1	68	63	129	37	1327	1839	1098	12.0	0.3	5.3	4
II-89-1	Orebody 1	134	137	108	90	1235	1159	717	7.1	0.3	9.7	4
II-89-15	Orebody 1	12	19	13	33	1062	1038	215	2.2	0.4	12.0	4
II-89-4	Orebody 1	48	38	62	25	1464	1048	2227	7.2	0.3	10.9	4
II-89-6	Orebody 1	175	159	190	65	1886	1139	964	14.3	0.4	6.6	4
II-W-11	Orebody 1	280	329	361	182	472	1010	635	3.1	0.3	9.8	4
II-W-13	Orebody 1	91	134	60	90	692	1014	1344	4.6	0.3	10.5	4
II-W-15	Orebody 1	42	61	56	53	539	707	1300	7.3	0.3	9.6	4
IIW-16	Orebody 1	175	180	221	115	796	924	722	1.3	0.4	6.6	4
II-W-19	Orebody 1	28	30	65	33	3236	1424	4214	13.6	0.3	9.3	4
II-W-21	Orebody 1	62	78	84	76	2238	1213	236	2.2	0.4	8.7	4
S89-3	Orebody 1	383	348	422	197	700	605	535	6.2	0.3	10.4	4
S89-4	Orebody 1	207	267	278	163	1768	1476	1125	10.2	0.3	7.5	4
S89-5	Orebody 1	179	153	256	129	781	1551	464	13.1	0.3	10.7	4
S89-6	Orebody 1	265	224	309	144	717	864	721	5.2	0.3	10.4	4
S89-7	Orebody 1	351	371	436	206	680	1057	492	3.3	0.4	10.6	4
S89-8	Orebody 1	340	319	408	298	927	736	451	2.0	0.3	12.0	4
050715-2	Orebody 2		10	9	4	430	479		4.5	0.4	7.2	1
JC02-7	Orebody 2		6	9	4	39	25	38	1.2	0.3	6.6	1
Y2-3	Orebody 2	554	522	532	170	1543	852	532	3.6	0.3	8.4	3
Y2-5	Orebody 2	97	116	87	68	822	1644	967	3.8	0.2	9.6	3
2-E-3	Orebody 2	17	16	28	11	108	244	154	7.2	0.3	9.1	4
2-E-4	Orebody 2	32	26	26	25	635	832	80	2.1	0.2	10.1	4
A2-32	Orebody 2	19	15	23	10	153	77	290	2.7	0.3	7.6	4
II-17	Orebody 2	32	26	41	22	838	603	171	3.1	0.3	8.3	4
II-26	Orebody 2	24	30	36	22	270	173	122	9.5	0.2	7.8	4
050712-6	Orebody 24		12	16	10	7645	3524		15.3	0.4	10.8	1
JC-2	Orebody 24		458	451	215	3652	2793		5.7	0.0	9.9	2
JC-20	Orebody 24		197	217	120	1458	1438		3.0	0.0	12.1	2
JC-23	Orebody 24		525	525	252	5925	2815		8.7	0.4	10.6	2
JC-26	Orebody 24		291	425	246	2549	3085		4.7	0.2	13.1	2
JC-3	Orebody 24		867	861	449	12228	6052		9.9	0.0	13.9	2
I-13	Orebody 24	426	557	590	308	8458	4829	2555	4.6	1.3	8.6	4
I-18	Orebody 24	107	124	112	55	4299	2581	1926	5.0	0.5	8.3	4
I-20	Orebody 24	221	289	344	138	5762	2765	972	5.6	0.6	9.2	4
I-21	Orebody 24	23	19	22	23	308	437	621	5.2	0.3	9.1	4
I-22	Orebody 24	209	195	162	115	8475	3889	2765	5.2	0.3	8.7	4
I-25	Orebody 24	100	95	132	39	2943	1356	659	3.2	0.7	9.0	4
I-27	Orebody 24	254	279	354	137	698	1484	632	2.6	0.4	8.2	4
I-29	Orebody 24	25	16	21	14	2273	3018	588	2.9	0.4	9.4	4

[a] Samples with Ir/(Ir + Ru)=0.3–0.7, Ir/(Ir + Rh)=0.4–0.8, Pt/(Pt + Pd)=0.3–0.7

Ref: 1, this work; 2, Song et al. 2006; 3, Yang et al. 2006; 4, Chai and Naldrett 1992a

Cu in the residual sulfide liquid. Other samples from the Jinchuan deposit are sub-parallel to each other, suggesting they are related to each other by different R-factors. Higher R-factors will result in higher concentrations for all PGE and Au in the sulfide liquids because they all have similarly high partition coefficients between sulfide liquid and magma.

Figure 9 illustrates variations of Pt and Pd with Ir for all samples listed in Table 2. Most samples show positive correlation between Pt and Ir (Fig. 9a) and between Pd and

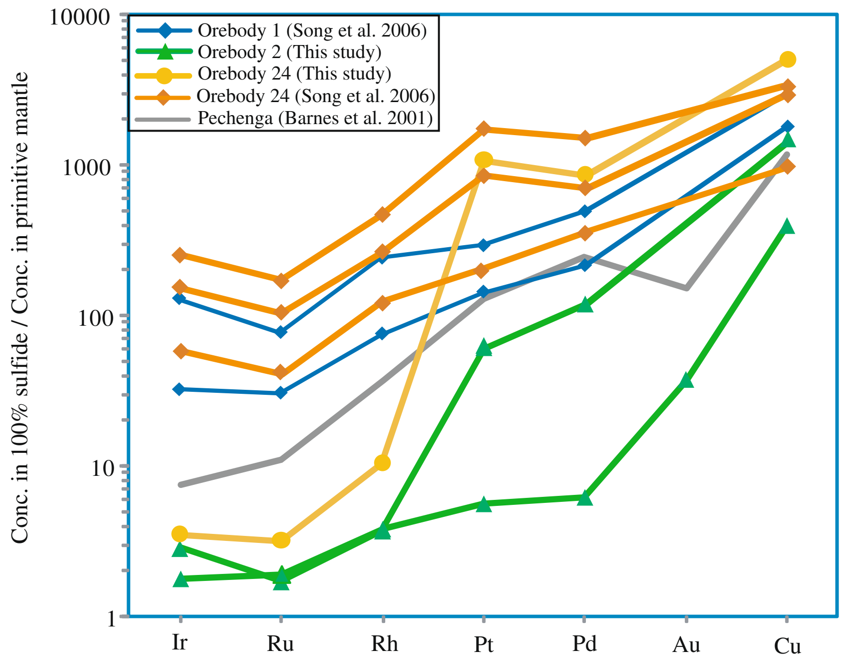

Fig. 8 Mantle-normalized patterns of PGE, Au, and Au for the Jinchuan deposit. Primitive mantle values of PGE + Au and Cu used in the normalization are from Barnes and Maier (1999) and McDonough and Sun (1995), respectively. The average values of disseminated sulfide ores for the Pechenga deposit is from Barnes et al. (2001)

Ir (Fig. 9b). A few samples from oreboby-1 and orebody-24 are displaced from the general trends. The positive correlations are consistent with variable R-factors during sulfide segregation, whereas the displacement can be explained by mss fractionation or hydrothermal redistribution. As we have filtered the samples using rather restricted ratios of Ir/(Ir + Ru), Ir/(Ir + Rh), and Pt/(Pt + Pd), we are quite confident that the co-variations of Pt and Pd vs Ir shown in the figure largely represent the results of magmatic processes.

We have attempted to model the magmatic processes using available experimental partition coefficients of Pt, Pd, and Ir between sulfide liquid and magma (Stone et al. 1990; Crocket et al. 1992; Bezmen et al. 1994; Peach et al. 1994;

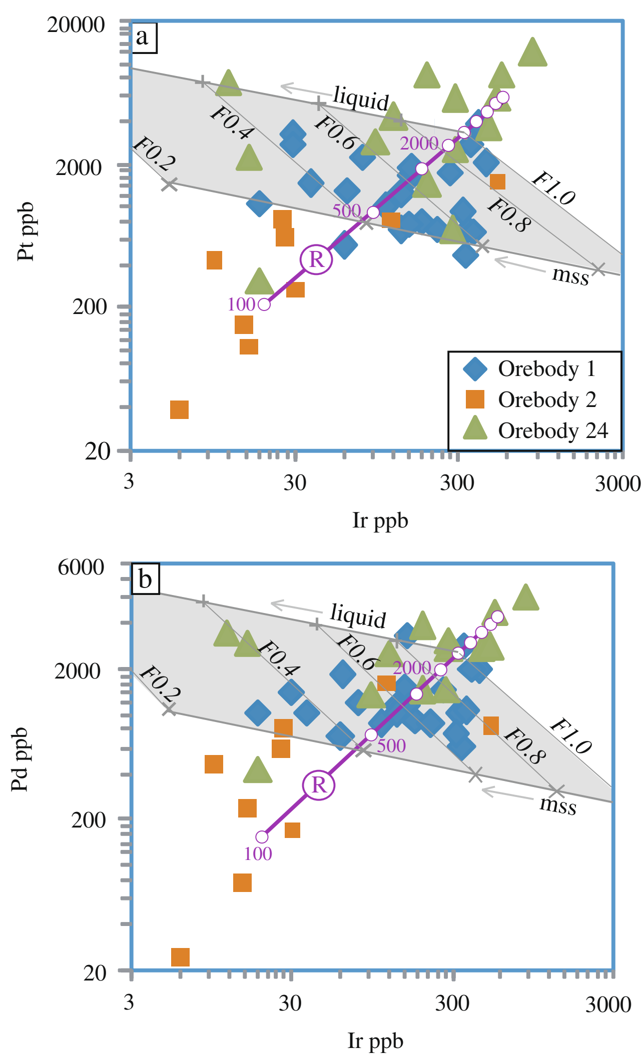

Fig. 9 Correlations between Pt and Ir (**a**), and Pd and Ir for samples ▶ with Ir/(Ir + Ru)=0.3–0.7, Ir/(Ir + Rh)=0.4–0.8, and Pt/(Pt + Pd)=0.3–0.7. These samples are believed to represent the least-altered samples from the Jinchuan deposit and may be used to evaluate primary processes. Element concentrations have been recalculated to 100% sulfide. *Line marked* with "R" represents sulfide liquids segregated from magma with variable *R*-factors. The *numbers beside the line* are the *R*-factors. The *shaded quasi-parallelograms* are defined by the compositions of mss and coexisting sulfide liquid. Numbered *F* is the fraction of remaining sulfide liquid. The sulfide/magma partition coefficients for the PGE are assumed to be 10^4. The mss/liquid partition coefficients for Ir and for both Pt and Pd are assumed to be 5 and 0.1, respectively. The partition coefficients used in our calculations are within the ranges of available experimental results. The concentrations of Ir=0.2 ppb, Pt=2 ppb, and Pd=1.5 ppb in the initial magma can well reproduce the overall trends of the selected samples from the Jinchuan deposit. Displacements from these trends can be explained by sulfide liquid fractionation

Fleet et al. 1999) and between mss and sulfide liquid (Fleet and Stone 1991; Fleet et al. 1993; Li et al. 1996; Barnes et al. 1997, Mungall et al. 2005). We used the mass balance equation by Campbell and Naldrett (1979) to calculate the R-factors and the conventional Rayleigh fraction equation for trace element enrichment (see Barnes and Lightfoot 2005) to simulate mss fractional crystallization. The sulfide/magma partition coefficients for Ir, Pt, and Pd are assumed to be 10^4, which are within the range of experimental values. The mss/liquid partition coefficients for Ir, Pt or Pd are 5 and 0.1, respectively. These values are within the range of experimental results.

Using these partition coefficients, we can well reproduce the compositional variations by assuming that the concentrations of Ir, Pt ,and Pd in the parental magma are 0.2, 2, and 1.5ppb, respectively (Fig. 9a,b). Our results indicate that some samples from orebody-1 and orebody-24 likely represent the products of sulfide liquid fractionation. The modeled R-factors for different ore bodies overlap, but in general they decrease from orebody-24 in the western part of the intrusion through orebody-1 in the central part of the intrusion to orebody-2 in the eastern part of the intrusion (Fig. 1b,c). Variable R-factors are required to explain the compositional variations within an individual ore body. This implies that the Jinchuan deposit was a dynamic system in which multiple sulfide liquids did not have a chance to homogenize, which is most consistent with multiple injections of sulfide-bearing magma from depth as proposed previously by Tang (1993), de Waal et al. (2004) and Li et al. (2004).

Parental magma composition

Chai and Naldrett (1992b) used the average Ni/(Ni + Cu) ratio of sulfide ores to estimate the parental magma composition for the Jinchuan intrusion. They concluded that the parental magma of the Jinchuan intrusion is of basaltic composition. The average value they used is slightly lower than the peak ratio of 0.7–0.8 for all samples from different studies (Fig. 7a). The peak value of the Jinchuan deposit is similar to the average value of 0.71 for the Pechenga deposit associated with picrites (Barnes et al. 2001). The PGE compositions of the Jinchuan deposit are also similar to that of the Pechenga deposit (Fig. 9).

However, the concentrations of Ir, Pt, and Pd in the parental magma of the Jinchuan intrusion inferred from the model results are significantly lower than the values in an undepleted picrite magma, which on average contains 0.8ppb Ir, 11ppb Pt, and 10ppb Pd according to Barnes and Lightfoot (2005). Depletion of PGE in the Jinchuan parental magma may have resulted from PGE retention at the mantle source or previous segregation of small amounts of sulfide liquid at depth.

Exploration implications

The indication of possible sulfide liquid fractionation in orebody-1 and orebody-24 (Fig. 9a,b) is important for underground exploration. Fractionated sulfide liquids can be substantially enriched in Pt, Pd, Au, and Cu. They are commonly generated after the solidification of host rocks. They can be expelled into footwall fractures hundreds of meters away from the basal contacts. A good example of this type of ore is the Deep Copper Zone of the Strathcona deposit at Sudbury, Canada (Li et al. 1992).

Conclusions

Key conclusions reached from this study are summarized below:

1. At least two thirds of disseminated and net-textured sulfide samples, and all massive sulfide samples from the Jinchuan deposit have been affected by postmagmatic hydrothermal alteration, resulting in decoupling of not only Pt and Pd but also Ru, Rh, and Ir.
2. Variable R-factors are required to explain PGE variations in each ore body as well as in the entire deposit, supporting the interpretation that multiple sulfide-bearing magmas from depth were involved in the formation of the Jinchuan deposit.
3. The contents of Ni, Cu, and PGE in the disseminated sulfide ores (recalculated to 100% sulfide) of the Jinchuan deposit are similar to those of the Pechenga deposit, suggesting the parental magmas of the Jinchuan and Pechenga intrusions are similar in PGE, Ni, and Cu concentrations.
4. Sulfide liquid fractionation likely took place in orebody-1 and orebody-24. The current effort to look for Cu-, Pt-, and Pd-rich sulfides in the footwall of these ore bodies is therefore justified.

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